

Students' Academic Analysis

Objective:

1. To analyze student academic performance
2. To identify patterns based on gender, subject, and attendance in order to support better teaching and learning decisions.

Questions Analyzed / Insights Built

1. Do male and female students perform differently?
2. Does attendance affect exam performance?
3. What is the average exam score by subject?
4. Which class level performs best overall?
5. Which subjects have the lowest average scores?

Student_ID	Gender	Class	Subject	Exam_Score	Attendance %	Term
STU001	Female	SS1	Maths	67	60	First
STU002	Male	SS3	Maths	78	70	First
STU003	Male	SS2	Maths	56	100	First
STU004	Female	SS3	English	45	60	First
STU005	Male	SS2	English	34	85	First
STU006	Female	SS1	Physics	80	70	First
STU007	Male	SS3	Maths	22	80	First
STU008	Male	SS3	English	58	80	First
STU009	Male	SS2	Chemistry	90	75	First
STU010	Female	SS1	Chemistry	77	65	First
STU011	Male	SS1	Biology	56	70	First
STU012	Female	SS1	Biology	75	75	First
STU013	Female	SS1	Biology	65	65	First
STU014	Female	SS3	Computer	83	60	First
STU015	Female	SS3	Computer	63	85	First
STU016	Male	SS2	Maths	62	90	First
STU017	Male	SS3	English	41	90	First
STU018	Female	SS2	Maths	45	90	First
STU019	Male	SS2	English	67	80	First
STU020	Male	SS2	Physics	34	95	First
STU021	Female	SS3	Chemistry	68	95	First
STU022	Male	SS1	Biology	95	100	First
STU023	Female	SS1	English	75	60	First
STU024	Male	SS1	English	64	60	First
STU025	Female	SS2	English	88	65	First
STU026	Female	SS2	English	55	75	First
STU027	Female	SS2	English	36	75	First
STU028	Female	SS3	English	73	80	First
STU029	Female	SS2	Biology	42	80	First
STU030	Male	SS1	Chemistry	46	80	First
STU031	Female	SS1	Physics	67	90	First
STU032	Male	SS1	Physics	87	90	First
STU033	Female	SS1	Biology	98	95	First

STU034	Male	SS1	Physics	76	90 First
STU035	Male	SS2	English	44	60 First
STU036	Male	SS3	Maths	58	65 First
STU037	Male	SS2	Maths	72	65 First
STU038	Female	SS2	Maths	71	65 First
STU039	Male	SS2	English	71	75 First
STU040	Female	SS3	English	38	75 First
STU041	Male	SS1	English	84	75 First
STU042	Female	SS2	Maths	55	75 First
STU043	Male	SS1	Chemistry	66	70 First
STU044	Female	SS2	Biology	70	70 First
STU045	Male	SS2	Biology	60	80 First
STU046	Female	SS3	Biology	86	85 First
STU047	Female	SS1	Biology	81	60 First
STU048	Female	SS1	Chemistry	28	65 First
STU049	Male	SS1	Physics	45	75 First
STU050	Male	SS1	Physics	67	70 First
STU051	Male	SS1	Physics	43	70 First
STU052	Female	SS1	Maths	57	75 First
STU053	Female	SS2	English	78	75 First
STU054	Male	SS2	English	88	90 First
STU055	Female	SS2	English	32	85 First
STU056	Male	SS3	English	43	85 First
STU057	Female	SS2	English	22	100 First
STU058	Female	SS1	English	56	75 First
STU059	Female	SS2	Maths	58	60 First
STU060	Female	SS3	Maths	67	65 First
STU061	Male	SS1	English	68	80 First
STU062	Female	SS1	Maths	36	80 First
STU063	Male	SS2	English	61	85 First
STU064	Male	SS1	Maths	60	85 First
STU065	Male	SS3	Chemistry	67	85 First
STU066	Male	SS2	Physics	43	70 First
STU067	Male	SS2	Computer	56	95 First

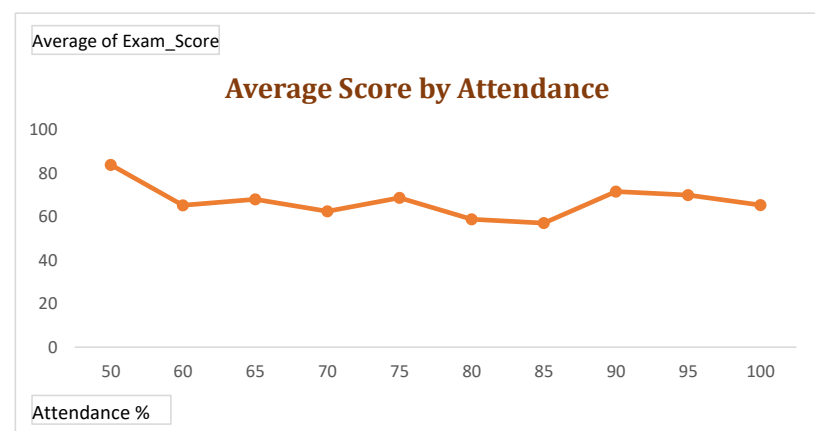
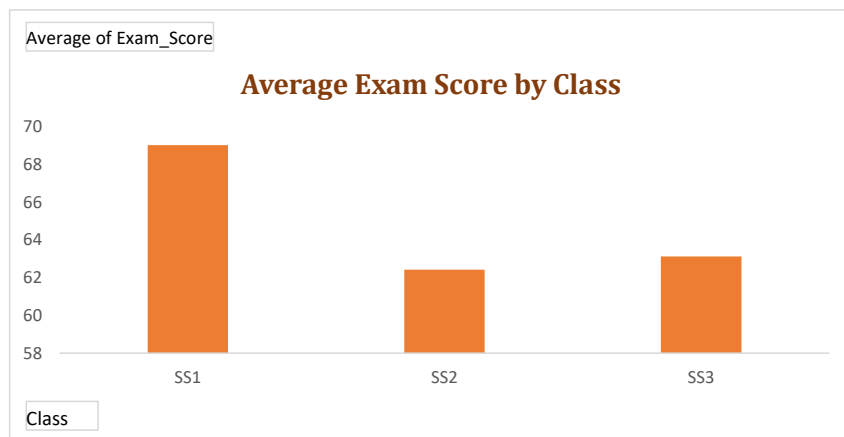
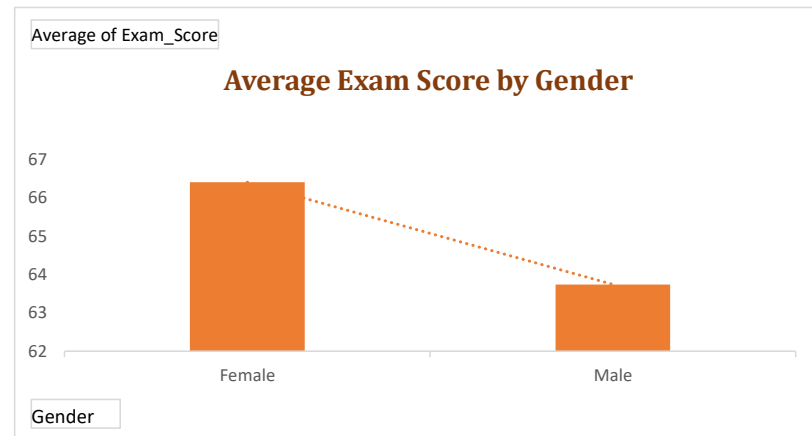
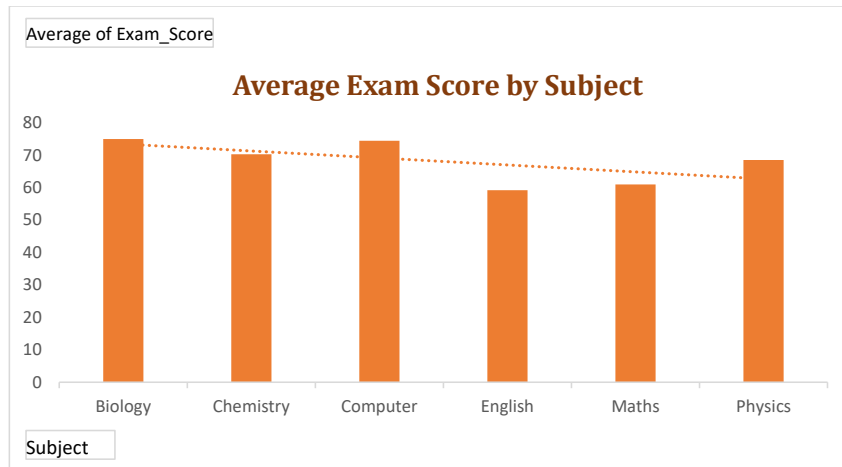
STU068	Male	SS3	Computer	89	95 First
STU069	Female	SS1	Computer	90	90 First
STU070	Male	SS1	Physics	91	75 First
STU071	Male	SS1	Physics	98	75 First
STU072	Female	SS1	Chemistry	78	95 First
STU073	Female	SS2	Chemistry	93	90 First
STU074	Female	SS2	Computer	92	90 First
STU075	Female	SS3	Computer	68	80 First
STU076	Male	SS2	Computer	48	85 First
STU077	Female	SS3	Maths	50	85 First
STU078	Male	SS1	Maths	52	80 First
STU079	Female	SS1	Maths	56	60 First
STU080	Female	SS1	Maths	87	65 First
STU081	Female	SS1	Maths	80	60 First
STU082	Male	SS2	English	82	75 First
STU083	Female	SS2	English	84	75 First
STU084	Female	SS3	English	47	85 First
STU085	Female	SS1	English	68	90 First
STU086	Female	SS1	Maths	66	85 First
STU087	Female	SS2	English	50	80 First
STU088	Female	SS3	English	70	70 First
STU089	Male	SS1	Maths	34	70 First
STU090	Male	SS2	English	46	85 First
STU091	Male	SS2	English	45	90 First
STU092	Male	SS2	Maths	78	90 First
STU093	Female	SS1	Computer	89	100 First
STU094	Female	SS3	Computer	67	95 First
STU095	Male	SS2	Physics	92	80 First
STU096	Female	SS2	Chemistry	90	80 First
STU097	Male	SS3	Biology	97	85 First
STU098	Female	SS1	Maths	84	50 First
STU099	Male	SS2	English	82	70 First
STU100	Female	SS3	Maths	74	75 First

Row Labels	Average of Exam_Score
Biology	75
Chemistry	70.3
Computer	74.5
English	59.21875
Maths	61
Physics	68.58333333
Grand Total	65.16

Row Labels	Average of Exam_Score
Female	66.41509434
Male	63.74468085
Grand Total	65.16

Row Labels	Average of Exam_Score
SS1	69.02564103
SS2	62.42105263
SS3	63.13043478
Grand Total	65.16

Row Labels	Average of Exam_Score
50	84
60	65.3
65	68.11111111
70	62.63636364
75	68.76470588
80	58.85714286
85	57.14285714
90	71.69230769
95	70
100	65.5
Grand Total	65.16



Key Findings

- Biology has the highest average exam score among the six subjects
- English recorded the lowest average performance by the students
- Female students outperformed the male students across most subjects having a percentage of 66%
- SS1 students has the overall performance in most subjects compared to SS2 and SS3
- Students with high attendance rates performed better and achieve higher exam scores compared to students with lower attendance rates

Recommendations

- Adequate academic support should be taken to condieration for subjects with lower grades especially English
- Teachers should encourage punctuality in classes for regular attendance, as it appears to positively influence academic performance.
- The teaching strategies that was adopted in Biology and English can be reviewed and adopted to other subjects with lower grades.

Conclusion

This analysis identified the patterns that target academic support in improving student performance. Data-driven insights such as the data gathered can support better curriculum structure, educational planning and decision-making.